1. What is PySpark DataFrame

PySpark DataFrame is a distributed collection of data organized into named columns.
It is similar to a table in a database and supports big data processing.

2. Key Features

Distributed processing
Schema-based structure
Optimized using Catalyst Optimizer
Supports SQL queries
Lazy evaluation

3. Creating DataFrame

From file: df = spark.read.csv('file.csv', header=True)
From list: df = spark.createDataFrame([(1,'A')], ['id','name'])

4. Schema

Defines structure of data (column name and datatype)
Example: df.printSchema()

5. Transformations

select() - Select columns
filter() - Filter rows
withColumn() - Add column
groupBy() - Group data
join() - Combine data

6. Actions

show() - Display data
count() - Count rows
collect() - Return all data
first() - First row

7. Column Operations

withColumn() - Add/modify column

withColumnRenamed() - Rename column

drop() - Remove column

8. Filtering Data

df.filter(df.age > 30)

df.filter((df.age > 25) & (df.salary > 5000))

9. Aggregations

df.groupBy('dept').count()

df.groupBy('dept').agg({'salary':'avg'})

10. Joins

Types: inner, left, right, outer

Example: df1.join(df2, 'id', 'inner')

11. Handling Null Values

dropna() - Remove nulls

fillna() - Replace nulls

12. Removing Duplicates

dropDuplicates()

distinct()

13. Sorting

orderBy('salary')

orderBy('salary', ascending=False)

14. Reading and Writing

Read: spark.read.csv(), spark.read.parquet()

Write: df.write.csv(), df.write.parquet()

15. Partitioning

repartition() - Increase/decrease partitions

coalesce() - Reduce partitions

16. Caching

cache() and persist() improve performance

17. Execution Plan

df.explain() shows execution plan

18. Spark SQL

df.createOrReplaceTempView('table')
spark.sql('SELECT * FROM table')

19. Lazy Evaluation

Transformations are not executed immediately
Execution happens only when action is called

20. Real-Time Understanding

DataFrame - Excel sheet
select() - Choose columns
filter() - Apply condition
groupBy() - Pivot
join() - VLOOKUP
write() - Save file